

2023 March Meeting Call for Nominations and Abstracts

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I have read the above and
understand the requirement
with regard to the Permissions
Agreement

Yes

Submission Type Oral Abstract

Symposia invited speakers can present, In-Person (Caesar's Forum in Las Vegas) ONLY. Please select the In-person option below.

You should have been instructed by your Unit organizer which option to select, Invited In-person (Caesar's Forum in Las Vegas) March 6-10, 2023 or Virtual (Online) March 20-22, 2023. Please select the appropriate option below.

You have the option of selecting Virtual (Online) March 20-22, 2023 or In-Person. (Caesar's Forum in Las Vegas) March 6-10, 2023. If you are not the presenter of this abstract, before proceeding, you must confirm the presentation format of the abstract with the speaker. This answer CANNOT be changed after the abstract submission deadline of October 20, 2022 for any reason.

Please note that there are two methods of presentation, IN-PERSON in Caesar's Forum Las Vegas and VIRTUAL(online).

If you are not the presenter of this abstract, before proceeding, you must confirm the presentation format of the abstract with the speaker. This answer CANNOT be changed after the abstract submission deadline of October 20, 2022 for any reason.

Presentation Method:

In-person (Caesar's Forum in Las Vegas) March 6-10, 2023

Sorting Category: 10 MAGNETISM (GMAG)

Sub-Category: 10.01.02 Emergent Properties of Complex Oxides Bulk, Thin Films, and Heterostructures (GMAG, DMP)

Would the speaker like to
volunteer as a session chair?

No

Abstract Title: Linear-in-temperature resistivity in PdCrO₂ from magneto-

Feedback and Support

elastic interactions

The kindred delafossite metals PdCrO₂ and PdCoO₂ exhibit a strikingly different temperature dependence of the in-plane resistivity at intermediate temperatures. The magnitude of the resistivity in the former displays a sublinear approach to a pronounced T-linear scaling. This difference, apparently linked to antiferromagnetic correlations in PdCrO₂, cannot be fully accounted for by conventional scattering mechanisms. Motivated by this puzzle, we consider a particular coupling of electrons to magneto-elastic (EME) interactions in a 3D system composed alternately stacked layers of itinerant electrons and antiferromagnetic Mott insulators. We show that EME interactions can explain the anomalous transport properties of PdCrO₂ and offer several experimental predictions to showcase the effects of the EME coupling.

Keyword 1: N/A

Keyword 2: N/A

Keyword 3: N/A

Category Type: Theoretical

Newsworthy Research? No

Order	Name	Role	Email	Affiliation	Presentation	Action
001	Juan F. Mendez-Valderrama	Speaker	jfm343@cornell.edu	Cornell University		Submitter
002	Evyatar Tulipman	Co-Author	evyatar.tulipman@weizmann.ac.il	Weizmann Institute of Science		
003	Debanjan Chowdhury	Co-Author	dc977@cornell.edu	Cornell University		
004	Erez Berg	Co-Author	erez.berg@weizmann.ac.il	Weizmann Institute of Science		

Feedback and Support